

I wrote
One Nested Query
A Teammate called
it Unreadable

ATLIQ HARDWARE GDB0041

MYSQL | Vishnu Ram Sachin | Data Analyst



Problem Statement

DAY 8 / 15

03/08/26

- Yesterday : net invoice sales , gross after pre invoice discount
- To reduce the Fiscal year pasted 5 times
- To enhance readability with adding 4 JOINS deep
- Finance wants **more deductions tomorrow**

The Trap ❌

- One nested block hides where each step begins and ends

5 copies of fiscal year

One miss edit , break the join



Requirment

DAY 8 / 15

03/08/26

Name your steps. Fiscal year written once



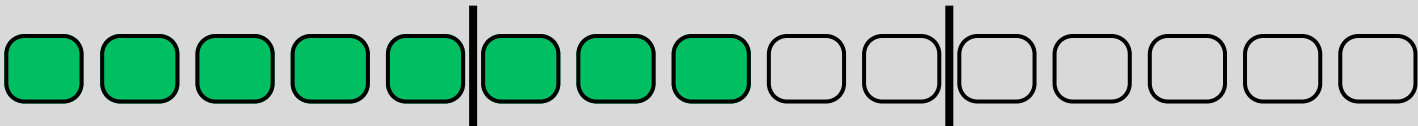
The Fix

Sales report > scope the raw sales here

Gross sales report > attach the price , sum to gross

pre invoice report > attach the deductions %

final SELECT > combine , compute net sales



Bonus Find

DAY 8 / 15

03/08/26

Readability has a direct cost: it's the time your next teammate spends understanding your logic before they can safely change it

1 Nested CTE

Burried Logic

----- new deduction means editing inside the block

1 Nested CTE

+ One job each

----- new deductions is one more CTE,



A Quick Analogy

DAY 8 / 15

03/08/26

Water purification plant - 4 steps

1. Sediment Filter

Removes dirt from the raw water
(sales_report)

2. Carbon Filter

Removes chlorine and odor
(gross_sales_report)

3. RO membrane

Removes dissolved salts
(pre_invoice_report)

4. UV Purifier

Kills bacteria , clean water out
(Final_SELECT)

One direction ,
One job per stage

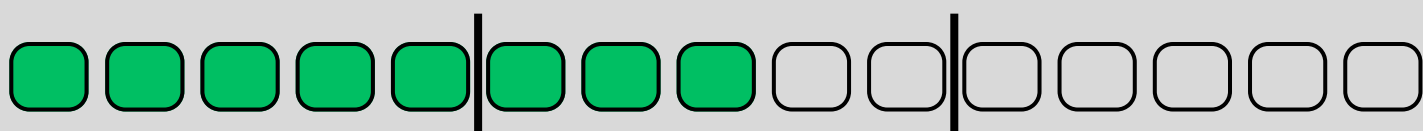


```
WITH sales_report AS (  
  SELECT fsm.date, fsm.fiscal_year, fsm.product_code,  
         fsm.customer_code, fsm.sold_quantity  
  FROM fact_sales_monthly fsm ),  
  
gross_sales_report AS (  
  SELECT  
    sr.customer_code, sr.fiscal_year,  
    ROUND(SUM(sr.sold_quantity * fgp.gross_price)/1000000,2) gross_sales_mln  
  FROM sales_report sr  
  JOIN fact_gross_price fgp  
  ON sr.product_code = fgp.product_code  
  AND sr.fiscal_year = fgp.fiscal_year  
  GROUP BY sr.customer_code, sr.fiscal_year) ,
```



```
pre_invoice_report AS (  
    SELECT  
        preivd.customer_code,  
        preivd.fiscal_year,  
        preivd.pre_invoice_discount_pct  
    FROM fact_pre_invoice_deductions preivd  
)
```

```
SELECT  
    gsr.customer_code,  
    gsr.gross_sales_mln,  
    ROUND(gsr.gross_sales_mln * (1 - pir.pre_invoice_discount_pct),2)  
net_invoice_sales  
FROM gross_sales_report gsr  
JOIN pre_invoice_report pir  
    ON pir.customer_code = gsr.customer_code  
    AND pir.fiscal_year = gsr.fiscal_year  
ORDER BY gsr.gross_sales_mln DESC;
```



Results

DAY 8 / 15

03/08/26

fact_sales_monthlyfact_gross_pricefact_pre_invoice_deductionsdim_productSQL File 8*SQL File 7* x

Limit to 50000 rows

```
1 • WITH sales_report AS (  
2   SELECT fsm.date, fsm.fiscal_year, fsm.product_code,  
3   fsm.customer_code, fsm.sold_quantity  
4   FROM fact_sales_monthly fsm ),  
5  
6   gross_sales_report AS (  
7   SELECT  
8     sr.customer_code, sr.fiscal_year,  
9     ROUND(SUM(sr.sold_quantity * fgp.gross_price)/1000000,2) gross_sales_mln  
10  FROM sales_report sr  
11  JOIN fact_gross_price fgp  
12  ON sr.product_code = fgp.product_code  
13  AND sr.fiscal_year = fgp.fiscal_year  
14  GROUP BY sr.customer_code, sr.fiscal_year) ,
```



Results

DAY 8 / 15

03/08/26

```
act_sales_monthly fact_gross_price dim_product fact_pre_invoice_deductions SQL File 8* SQL File 7* x
Limit to 50000 rows
16 pre_invoice_report AS (
17     SELECT
18         preivd.customer_code,
19         preivd.fiscal_year,
20         preivd.pre_invoice_discount_pct
21     FROM fact_pre_invoice_deductions preivd
22 )
23
24 SELECT
25     gsr.customer_code,
26     gsr.gross_sales_mln,
27     ROUND(gsr.gross_sales_mln * (1 - pir.pre_invoice_discount_pct),2) net_invoice_sales
28 FROM gross_sales_report gsr
29 JOIN pre_invoice_report pir
30     ON pir.customer_code = gsr.customer_code
31     AND pir.fiscal_year = gsr.fiscal_year
32 ORDER BY gsr.gross_sales_mln DESC;
```



Results

DAY 8 / 15

03/08/26

fact_sales_monthlyfact_gross_pricefact_pre_invoice_deductionsSQL File 8SQL File 7

Limit to 50000 rows

16

pre_invoice_report AS (

17

SELECT

18

preivd.customer_code,

19

preivd.fiscal_year,

20

preivd.pre_invoice_discount_pct

21

FROM fact_pre_invoice_deductions preivd

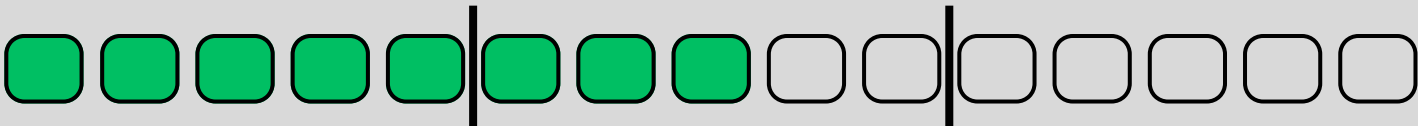
Result Grid

Filter Rows:

Export:

Wrap Cell Content:

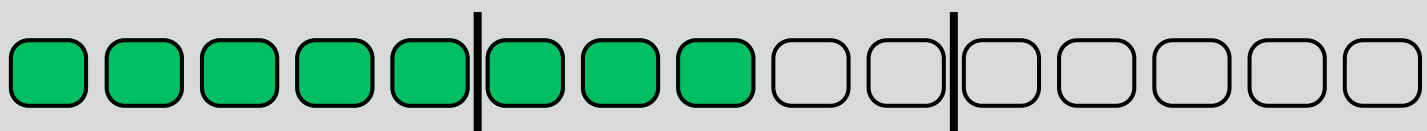
	customer_code	gross_sales_mln	net_invoice_sales
▶	80007196	103.55	74.33
	80007195	99.78	70.55
	90002008	84.16	59.65
	80001019	73.71	59.76
	90002009	57.59	46.68
	90002004	54.03	42.67
	90002016	53.77	37.52
	90002012	53.37	39.29
	80006155	53.32	37.65
	90002015	51.24	40.19
	90002007	50.31	37.23
	90002011	49.70	40.03
	90002003	49.26	35.14
	80007196	48.69	34.96
	80007195	48.25	35.62





DAY 8 / 15

03/08/26



I Vishnu Ram Sachin I Data Analyst

SQL THROUGH ONE DATASET CHALLENGE